Lecture 2 - Generative Learning Algorithm

So far we have learn the 2-class classification task, but what we have done before is the **discriminative algorithm**, as we learn from the features and provide the prediction for that class. So now, we will move on to another algorithm, called **generative algorithm**

	Generative - learn class by class	**Discriminative - learn from that features**
Learn what?	$p(x\|\|y)$ and $p(y)$	$p(y\|\|x)$ directly
Example	GDA, Naive Bayes	Logistic Regression

1. Gaussian discriminant analysis

1.1 The multivariate normal distribution

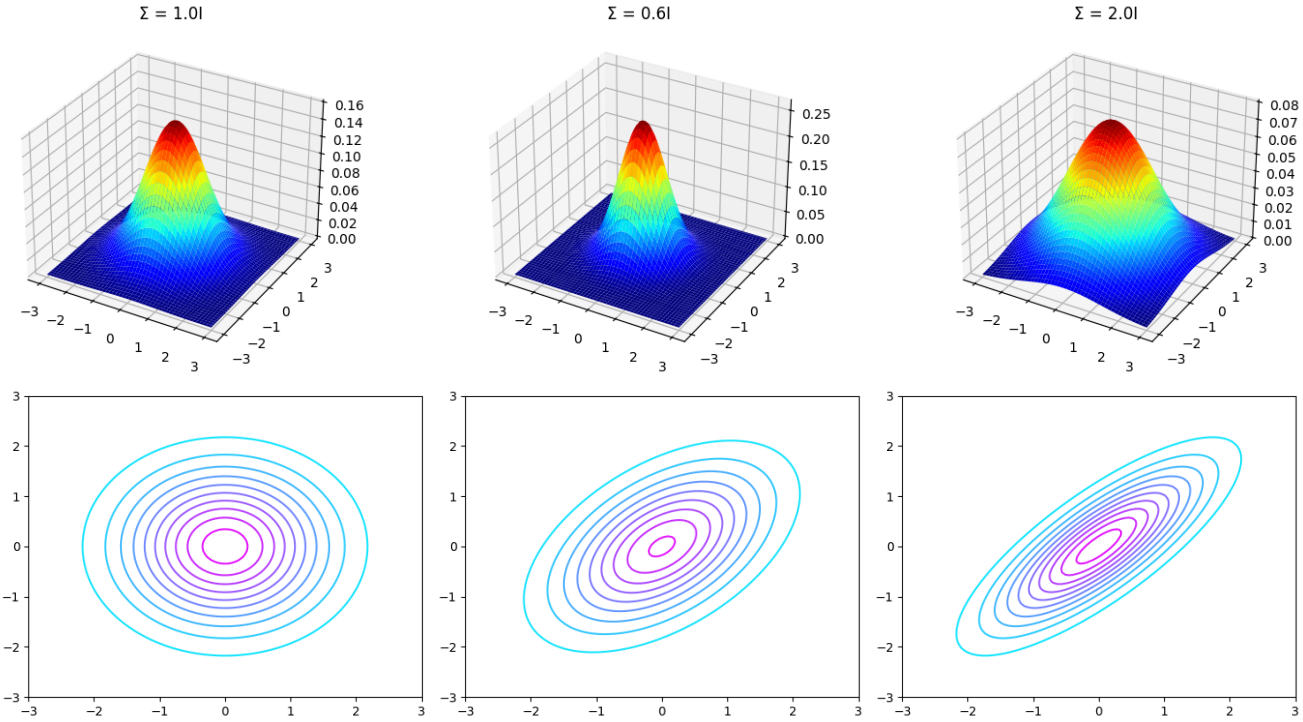
The multivariate normal distribution in n-dimensions, also called the multivariate Gaussian distribution, is parameterized by a **mean vector** $\mu \in \mathbb{R}^n$ and a **covariance matrix** $\Sigma \in \mathbb{R}^{n \times n}$, also written "$\mathcal{N}(\mu, \Sigma)$" and its density is given by:

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right).$$

where $|\Sigma|$ denotes the determinant of the matrix Σ .

The **covariance** of a vector-valued random variable Z is defined as

$$\mathrm{Cov}(Z) = E[(Z - E[Z])(Z - E[Z])^T]$$



1.2 The Gaussian Discriminant Analysis model

When we have a classification problem in which the input features x are continuous-valued random variables, we can then use the Gaussian Discriminant Analysis (GDA) model, which models $p(x|y)$ using a multivariate normal distribution. The model is:

$$\begin{aligned} y &\sim \text{Bernoulli}(\phi) \\ x|y=0 &\sim \mathcal{N}(\mu_0, \Sigma) \\ x|y=1 &\sim \mathcal{N}(\mu_1, \Sigma) \end{aligned}$$

Writing out the distributions, this is:

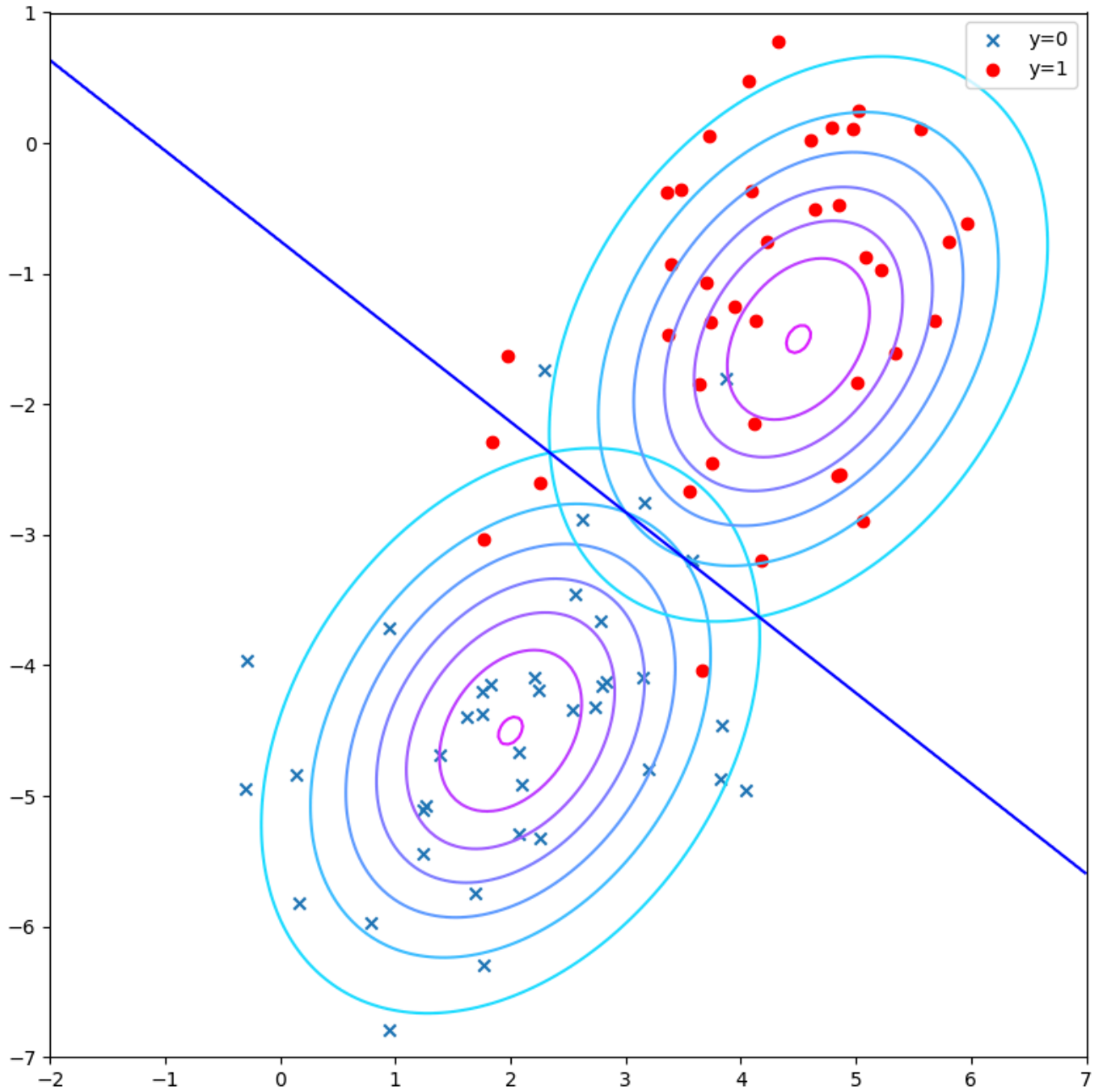
$$\begin{aligned} p(y) &= \phi^y(1-\phi)^{1-y} \\ p(x|y=0) &= \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_0)^T\Sigma^{-1}(x-\mu_0)\right) \\ p(x|y=1) &= \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right) \end{aligned}$$

Here, the parameters of our model are ϕ, Σ, μ_0 and μ_1. (Note that while there're two different mean vectors μ_0 and μ_1, this model is usually applied using only one covariance matrix Σ.) The log-likelihood of the data is given by

$$\begin{aligned} \ell(\phi, \mu_0, \mu_1, \Sigma) &= \log \prod_{i=1}^{m} p(x^{(i)}, y^{(i)}; \phi, \mu_0, \mu_1, \Sigma) \\ &= \log \prod_{i=1}^{m} p(x^{(i)}|y^{(i)}; \mu_0, \mu_1, \Sigma) p(y^{(i)}; \phi). \end{aligned}$$

By maximizing ℓ with respect to the parameters, we find the maximum likelihood estimate of the parameters (see problem set 1) to be:

$$
\begin{aligned}
\phi &= \frac{1}{m}\sum_{i=1}^{m} 1\{y^{(i)} = 1\} \\
\mu_0 &= \frac{\sum_{i=1}^{m} 1\{y^{(i)} = 0\}x^{(i)}}{\sum_{i=1}^{m} 1\{y^{(i)} = 0\}} \\
\mu_1 &= \frac{\sum_{i=1}^{m} 1\{y^{(i)} = 1\}x^{(i)}}{\sum_{i=1}^{m} 1\{y^{(i)} = 1\}} \\
\Sigma &= \frac{1}{m}\sum_{i=1}^{m}(x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T.
\end{aligned}
$$



2. Bayes

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

From the formulation above, the probability of event A when B is known will depend on 3 factors:

- $P(B|A)$ the probability of B happening when A is known to have occurred.
- $P(A)$ the probability of A happening on its own, regardless of B. It is also called **prior probability (xác suất tiên nghiệm)**
- $P(B)$ the probability of B happening on its own, regardless of A. It is called **normalising constant**

There is also a form of Bayes' theorem for continuous distributions. We have the probability density function

$$f(x|y) = \frac{f(y|x)f(x)}{f(y)}$$

By the way, a property of statistical probability is

$$\begin{aligned} p(x) &= \sum_y p(x|y)p(y) \\ &= p(x|y=1)p(y=1) + p(x|y=0)p(y=0) \end{aligned}$$

3. Naive Bayes

To model $p(x|y)$, we will therefore make a very strong assumption: the x_i are conditionally independent of given y , which means that

$$p(x_1, \ldots, x_{50000}|y) = \Pi_{i=1}^n p(x_i|y), \quad i = 1, \ldots, 50000$$

Having fit all these parameters, to make a prediction on a new example with features x, we then simply calculate

$$\begin{aligned} p(y=1|x) &= \frac{p(x|y=1)p(y=1)}{p(x)} \\ &= \frac{\left(\prod_{i=1}^n p(x_i|y=1)\right) p(y=1)}{\left(\prod_{i=1}^n p(x_i|y=1)\right) p(y=1) + \left(\prod_{i=1}^n p(x_i|y=0)\right) p(y=0)}, \end{aligned}$$

and pick whichever class has the higher posterior probability.